PACE Yourself

Pause, Attune, Care, Engage: A Practice for Infection-Associated Chronic Illness

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Chapter 1

PACE Yourself

1.0.1 Pause, Attune, Care, Engage: A Practice for Infection-Associated Chronic Illness

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First edition.

1.1 Who this book is for

This book is for the reader who has already read *Understanding Infection-Associated Chronic Illness* – or who does not need to, because they already live it. You know what post-exertional malaise feels like. You have already had the conversation where a test came back normal and nobody could say why you still can't get through a day. That first book gave you the map: what is established, what is a well-supported hypothesis, what is genuinely contested. This book gives you the practice built on that map – four verbs, run daily, for staying inside the energy you actually have.

1.2 The word PACE

"PACE" was the name of a 2011 UK clinical trial of graded exercise therapy and cognitive behavioral therapy for chronic fatigue syndrome. Its reported recovery rates shaped clinical guidance for a decade. It also had real methodological problems, documented in the published record, not argued here as advocacy: outcome measures changed partway through the trial, the thresholds for counting a participant "recovered" overlapped with the thresholds for enrolling them, and the trial never

measured post-exertional malaise, the symptom this book is built around. In 2021 the UK's National Institute for Health and Care Excellence set aside its old graded-exercise recommendation for myalgic encephalomyelitis/chronic fatigue syndrome and named pacing instead (National Institute for Health and Care Excellence 2021).

This book takes the same four letters back for the opposite idea. PACE Yourself is not that trial, and it is not a graded, scheduled increase in activity regardless of how you feel. It is pacing – the strategy the PACE trial's approach ignored, built on staying inside the energy you actually have rather than pushing through to expand it.

1.3 How the four verbs work

Pause means stopping before the limit, not at it. Post-exertional malaise arrives twelve to seventy-two hours after the exertion that triggers it, so waiting to feel the edge means you have already crossed it; Chapter 1 shows you how to set a ceiling from your own history instead.

Attune means reading your own baseline, never a population threshold. Chapter 2 shows you how to notice when your heart rate, sleep, or symptoms drift from what is ordinary for you, and how to treat every rule of thumb – including the heart-rate heuristics already in circulation – as a heuristic, never a diagnosis.

Care means relation-repair, never extraction: no upsell, no guilt, no streak punishment, no diagnosis sold as certainty. Chapter 3 covers the practical work inside your envelope – orthostatic management, sleep, hydration, sensory load – each named honestly by what its evidence actually supports.

Engage means re-entering activity on your body's terms, never a calendar's. Chapter 4 shows you how to advance only once you have fully recovered from your last exertion, the mechanistic opposite of a graded, scheduled increase.

1.4 How this book is organized

Chapter 1, Pause, covers why the felt limit arrives too late to be useful and how to set your ceiling ahead of it. Chapter 2, Attune, covers reading your own baseline across your history and the honest evidence state of the biometric tools available to help you do it. Chapter 3, Care, covers the daily practices that repair rather than extract inside the envelope Chapters 1 and 2 help you find. Chapter 4, Engage, covers re-entering activity on symptom-contingent terms and why that differs from graded exercise. Chapter 5, The Loop, brings the four verbs together into the shape of a day – and of the second day, when the loop is audited – and how to re-estimate your envelope weekly.

1.5 A note on honesty

The controlled evidence for pacing itself is still limited. As Chapter 4 of *Understanding Infection-Associated Chronic Illness* lays out, most of what supports pacing is patient experience, mechanistic reasoning, and observational data, not the large randomized trials that would say with confidence how much it helps or for whom. This book teaches pacing anyway, for reasons stated plainly: the mechanistic case is sound, patient experience across decades points the same direction, and the documented harm of the opposite approach – pushing through regardless of symptoms – is better evidenced than pacing's benefit is. Where a step here rests on a single small study or an unvalidated heuristic rather than an established finding, it says so, the same way the first book did.

1.6 A note on citations

Where this book cites a specific study, you'll see a bracketed marker like (National Institute for Health and Care Excellence 2021) after the sentence; the full citation appears in the references at the end of the chapter it belongs to. Most of this book's claims about the underlying science were already cited once, in *Understanding Infection-Associated Chronic Illness*, and this book points you back there rather than repeating the same citations twice. You don't need to interrupt your reading to check any of it, but it's there whenever you want to see where a claim came from.

Chapter 2

Chapter 1: Pause

Pause is the first move of PACE Yourself, and it is the one that goes against everything you have probably been taught about pushing through. Pause means stopping before the limit, not at it. Not when you feel bad. Not when your body finally objects. Before.

That sounds backward if you have never lived with post-exertional malaise (PEM) — the delayed, disproportionate symptom flare that follows exertion in myalgic encephalomyelitis, long COVID, and the other infection-associated chronic illnesses. If you have, you already know why waiting for the signal is the wrong plan: the signal doesn't come on time. Book 1 walked through what PEM is and the honest evidence for pacing as a strategy against it. This chapter doesn't repeat that ground. It takes one piece of that science — the delay — and asks what it actually requires you to do, today, before you feel anything at all.

2.1 Why the moment cannot be felt

Ordinary fatigue gives you a warning you can act on. You get tired, you notice, you sit down. The feedback loop is fast enough to steer by.

Post-exertional malaise breaks that loop. A short walk, half an hour of housework, a stretch of concentrated thinking, even a hard conversation — something modest enough that you might not register it as "exertion" at all — can produce a severe flare 12 to 72 hours later. Not a bad afternoon. A flare that lasts days to weeks, not hours (Stussman et al. 2020). In long COVID specifically, more than half of people report experiencing this kind of delayed crash within the first few months after infection (An et al. 2024). This is not a rare or unusual reaction; for a large share of people with these illnesses, it is the baseline pattern.

The delay is exactly what makes "listen to your body" incomplete advice here. By the time your body tells you something was too much, the exertion that caused it is a day or two in the past, and the choice you'd want to make differently already happened. You cannot feel the ceiling at the moment you cross it, because the cost hasn't arrived yet.

There is a second piece to this, and it changes how you should picture your own capacity. It is tempting to think of your energy as a tank you can run down and then have to refill — get a little more tired with every task, until you're empty. The

research on exercise capacity in myalgic encephalomyelitis points to something different: capacity behaves less like a tank and more like a rate — a ceiling on how much your body can do at once, not a reserve that simply depletes. Two studies using a repeated exercise test on consecutive days found that people with myalgic encephalomyelitis showed measurably lower performance on the second day than the first — a pattern healthy people and people with ordinary deconditioning do not show (Keller, Pryor, and Giloteaux 2014; Snell et al. 2013). A larger 2026 study using the same protocol did not find that same second-day decline, so this specific test result is genuinely unsettled in the research right now (Mancini et al. 2026). What is not in dispute across these studies is the basic shape of the problem they are all trying to measure: people are hitting a functional ceiling at levels of exertion that would barely register for someone without the illness. If capacity is a rate ceiling rather than a stock, the failure doesn't show up as "ran out late in the day." It shows up as failing at low intensity, sooner than seems like it should be possible — which is exactly what pacing has to plan around.

What this means today: you cannot rely on real-time symptoms to tell you when to stop, because the bill for today's activity may not arrive until the day after tomorrow. The plan has to be set before you start, using what you already know about your own pattern — not decided in the moment by how you currently feel.

2.2 The forms of pausing

Pausing is not one thing. It takes several shapes, and most people end up using more than one at a time.

Pre-emptive rest blocks are scheduled stops built into your day whether or not you feel you need them yet — twenty minutes lying down after breakfast, a rest period built into the middle of an errand, a fixed stopping point on a task before it's finished. The point of a pre-emptive block is that it doesn't wait for a symptom to justify it. You take it because it's the plan, not because you've earned it by suffering first.

Time-boxed activity puts a limit on duration rather than on how the activity feels. Instead of "I'll fold laundry until I'm done," it's "I'll fold laundry for fifteen minutes, then stop, regardless of how much is left." This matters because how a task feels in the first fifteen minutes is a poor predictor of what it will cost you two days later — the feedback you'd normally use to decide "a little more is fine" isn't trustworthy here.

Cognitive and emotional exertion count too. This is the part people most often miss, because it doesn't look like exertion. Concentrating hard at a screen, following a fast conversation, doing your taxes, sitting with a difficult phone call, holding your composure through a stressful meeting — all of these can trigger

the same delayed flare as physical activity. If your pacing plan only accounts for physical movement, you can do everything "right" physically and still get flattened by a day that was mentally or emotionally demanding. A pause has to budget for all three: body, mind, and feeling.

What this means today: look at your day for cognitive and emotional load, not just physical tasks. A pause plan that only tracks steps and stairs is only pacing a third of what actually costs you.

2.3 The heart-rate ceiling: a rule of thumb, not a rule

Because the moment cannot be felt, a lot of people look for a number instead — something objective to watch instead of an unreliable internal signal. The most common version of this is a heart-rate ceiling: stay below a certain heart rate during activity, and stop or slow down if you go above it.

The best-known version of this is the Workwell Foundation's rule of thumb: keep your heart rate below your resting heart rate plus 15 beats per minute during activity. It is worth knowing this rule exists, because it circulates widely in patient communities and clinical practice, and it is a genuinely useful anchor for many people. It is also worth knowing, plainly, what it is and isn't: it is clinical practice guidance, developed by a rehabilitation practice with real experience treating this illness — not a threshold that has been tested in a dedicated study and shown to predict or prevent post-exertional malaise. The same is true of the other common rule of thumb, staying under a fixed percentage of your age-predicted maximum heart rate: it is a reasonable, cautious starting point, not a validated instrument.

That distinction matters in a specific way. A heart-rate ceiling can be a useful, honest tool — a number you can watch in real time, when the delayed nature of PEM makes real-time symptoms useless. But it should be held the way you'd hold any rule of thumb: as a starting estimate to test against your own experience and adjust, not as a scientifically proven line that guarantees safety below it and danger above it. Nobody has published the study that would let anyone say that with confidence, for this particular number, yet.

What this means today: if you use a heart-rate ceiling, use it as a starting guess you refine from your own results — not as a promise. Watch what happens after you stay under it, and after you go over it, and let your own pattern, not the number alone, tell you where your real ceiling sits.

2.4 The calendar is not the signal

There is a second kind of ceiling that deserves the same skepticism, and it's more familiar because it's built into how most rehabilitation and fitness advice works:

the calendar. Do a little more this week than last week. Increase duration or intensity on a fixed schedule, regardless of how you're doing, on the theory that steady, gradual increases will build your capacity back up.

This approach has a name — graded exercise therapy — and for many conditions it works exactly as intended: deconditioning responds to gradual reintroduction of activity. For post-exertional malaise, it is the wrong tool, because it is built on the assumption pacing exists to reject: that today's tolerable-seeming exertion is safe to build on, when the actual cost of today's exertion won't be visible until the day after tomorrow. A schedule that increases activity regardless of symptom response will, sooner or later, increase activity past a ceiling it cannot see coming.

This is not a new or untested claim. "PACE" was the name of a large 2011 UK trial that tested graded exercise therapy and cognitive behavioral therapy for myalgic encephalomyelitis / chronic fatigue syndrome. The trial had real, documented methodological problems: its outcome measures and recovery thresholds were changed partway through the trial, the thresholds it used to count someone as "recovered" overlapped with the criteria used to let someone into the trial in the first place, and it did not measure post-exertional malaise at all. The UK's National Institute for Health and Care Excellence reviewed this record and, in its 2021 guideline, withdrew its earlier recommendation for graded-exercise-based treatment, replacing it with a caution that pacing — not a fixed schedule — may be more appropriate (National Institute for Health and Care Excellence 2021). That is published record, not advocacy: the trial's own design choices are why its results are no longer treated as a reliable guide to what to do.

PACE Yourself takes its name from the same three letters on purpose, and does the opposite of what that trial's approach did. It is pacing — the strategy that trial's schedule-based approach set aside — reclaimed as the name for the thing patients actually do to manage this illness. The calendar is not the signal here. Your own recent history is.

What this means today: if a plan tells you to do more this week than last week simply because a week has passed, that plan is answering to the wrong authority. Any increase in activity should answer to how you actually did after the last one — not to the date.

2.5 Pausing on a good day

The hardest pause to take is the one on a day when you feel fine. If your envelope for a task is twenty minutes and you're at minute fifteen feeling completely okay, every instinct says keep going — you're doing well, why waste it?

This is the discipline pausing actually asks of you: stopping at fifteen when you could plausibly keep going to twenty, or beyond. Not because fifteen is a magic

number, but because the whole point of staying inside your envelope is to never find out, the hard way, where the edge of it actually was today. A good day is not evidence your envelope has grown. It is one data point, and pushing past it to test that theory is how the two-day-later crash gets set up.

This is also where pacing differs most sharply from an ordinary "listen to your body and rest when needed" approach. Ordinary fatigue rewards pushing a little on a good day — you'll feel it if you overdo it, and you can back off. Post-exertional malaise punishes that same instinct on a delay long enough that the good day and the bad day never feel connected to each other. Stopping early on a day that felt easy is not caution for its own sake. It is the only way the delay in this illness can be planned around at all.

What this means today: notice the moment you catch yourself thinking "I feel good, I could keep going." That thought is the cue to stop, not the reason to continue.

2.6 What to do today

You don't need a perfect system to start pausing. You need a few honest decisions, made in advance, before you're relying on a symptom that will arrive too late to help you.

Pick one recurring task in your day — a chore, a work block, a walk — and set a time limit for it before you start, shorter than you think you need. Notice, separately from physical activity, one thing today that costs you cognitively or emotionally, and give it a boundary the same way. If you use a heart-rate number, write down what you're using it for and treat it as a guess you're testing, not a rule you've proven. And the next time a task is going well and you feel the pull to keep going a little longer, treat that feeling itself as the stop sign.

None of this requires knowing your exact envelope yet. Attune, the next chapter, is about how you find that out from your own data over time. Pause is what buys you the room to look: stop early enough, consistently enough, that the two-day-later flare has nothing to tell you, because you never crossed the line it would have punished.

Chapter 3

Chapter 2: Attune

3.1 Your own history is the instrument

Somewhere in the last few months, you have probably been handed a number that was supposed to mean something and didn't. A resting heart rate flagged "elevated" by a smartwatch that has no idea what your resting heart rate actually runs. A lab value inside the reference range, delivered as reassurance, on a day you could barely stand at the sink. Population thresholds are built for populations. You are not a population. You are one person with one history, and the only baseline that tells you anything true about today is the one built from your own yesterday.

Attune is the practice of reading that history. Not a single measurement, but the pattern behind it: your resting heart rate over the last month, not just this morning's reading; how last night's sleep compares with your usual nights, not with eight hours of textbook sleep; how today's fatigue and brain fog sit against the run of days before it. A population threshold can tell you whether you are unusual compared with strangers. Only your own rolling history can tell you whether today is unusual compared with you.

This matters more, not less, in infection-associated chronic illness, because the whole point of pacing is staying inside a capacity that is specific to you and that moves — week to week, sometimes day to day. A "normal" resting heart rate for the general population is a wide range that says nothing about whether 76 beats per minute is an ordinary Tuesday for you or the first sign that today is not going to be an ordinary Tuesday. The instrument that answers that question is not a chart in a textbook. It's the log of your own days.

What this means today: if you're tracking anything at all — heart rate, sleep, symptoms, steps — the number that matters is not today's reading against a general chart. It's today's reading against your own last few weeks.

3.2 What the tools can honestly tell you

There is a real and growing set of tools that read heart rate, heart-rate variability (HRV — a measure of the tiny variation in time between heartbeats), sleep, and activity, and turn them into pacing guidance. Some of them are worth using. None

of them can do what a headline about them sometimes implies. Here is the honest version, plainly stated, of what the current evidence actually supports.

Start with the two-day exercise test, the closest thing this field has to an objective, repeatable signature of post-exertional malaise: a maximal exercise test, repeated twenty-four hours later, to see whether the body reproduces its own day-one performance. Two studies found that people with ME/CFS characteristically could not — their oxygen uptake, heart rate, and workload at peak effort all dropped measurably on day two, in a pattern healthy people and people with ordinary deconditioning do not show (Keller, Pryor, and Giloteaux 2014; Snell et al. 2013). That looked, for a while, like a durable biological marker. Then a larger 2026 study using the same protocol, with both patients and healthy controls, found no such decline at all, and its authors said so directly: their data did not support using the two-day test to define post-exertional malaise or disability (Mancini et al. 2026). This is not one study being wrong and another being right. It is a live, unresolved disagreement in the published record, and no tool built on top of it — biometric or otherwise — can honestly claim to detect post-exertional malaise from a two-day decline. The most it can do is describe what happened on day two.

Now the newer, smaller signals. A 2025 study built a model that predicted next-day fatigue from HRV, sleep, and activity data with real accuracy — but in sixty-eight people with post-COVID fatigue, with no separate group used to check whether the model still works on people it wasn't built from (Aboagye et al. 2025). A 2024 feasibility study had thirteen people with long COVID practice slow breathing with HRV feedback for four weeks and found real, statistically meaningful improvement in autonomic symptoms and disability scores — again, thirteen people, no comparison group, and the authors' own conclusion was that the small sample limits how far the result generalizes (Corrado et al. 2024). Both are genuine, promising, single-cohort findings. Neither has been confirmed by an independent team, in a different group of people, using a different device. That doesn't make them worthless. It makes them early.

And then there are the rules of thumb already circulating in clinics and patient communities — most commonly, stay below your resting heart rate plus fifteen beats per minute during activity. This heuristic comes from a rehabilitation practice's clinical experience, and it may well be useful. But a direct search of the published literature for a study that tested this specific threshold against post-exertional-malaise outcomes turns up nothing. It is a heuristic in wide informal use, not a validated instrument, and it should be offered to you as exactly that — never with the authority of a tested measurement.

What this means today: any tool, including the pacing calculator this book will point you to, can honestly do three things with your biometric data — show it to you, notice when it drifts from your own pattern, and offer a heuristic labeled as

a heuristic. It cannot diagnose post-exertional malaise from a heart-rate reading, and you should be skeptical of anything that claims it can.

3.3 The second day is the reading that matters

If there is one habit worth building out of everything in this chapter, it's this: check tomorrow, and the day after, not just today. Post-exertional malaise characteristically arrives twelve to seventy-two hours after the exertion that triggers it and then lasts for days (Stussman et al. 2020). That delay is the whole reason a felt-in-the-moment approach fails — Chapter 1 covered why the pause has to be set in advance — and it's also why attunement can't stop at "how do I feel right now." The read that actually tells you whether yesterday's activity was inside your envelope is not yesterday. It's the day after tomorrow.

In practice, this means holding two days in your head at once: what you did, and what happened after. Did Tuesday's grocery run get followed by an ordinary Wednesday, or did Thursday arrive with the fog and the ache that mean the run cost more than it looked like it cost at the time? You cannot answer that on Tuesday. You can only answer it by checking back. Over weeks, that habit — activity, then a look at the second day, then a look at the day after that — is what actually builds an accurate picture of your envelope. A single day's data point tells you almost nothing. The relationship between an activity and what follows it, tracked over time, tells you a great deal.

What this means today: when you log an activity, set a reminder — a note, an alarm, whatever works — to check back in two and three days. That delayed check is worth more than anything you can feel in the moment.

3.4 Watching relations, not one number

There's a reason attunement is about more than any single reading, and it comes from how your body's stress response actually works. Two systems answer the same demand — standing up, walking to the mailbox, a hard conversation — on two different clocks. One is fast and neural: it fires in seconds, raises your heart rate, and gets your blood moving. The other is slow and hormonal: it takes fifteen to thirty minutes to ramp up, and its job is to fund the fast system's demand, hold your blood vessels responsive to it, and eventually shut the whole response down. In a body where these two arms are working together, that's invisible — it just works. Current research on ME/CFS proposes that in this illness, the two arms come apart: the fast system fires normally, but the slow system that's supposed to fund and end it does not keep up. That is a well-supported hypothesis, not yet a settled fact, but it explains a lot that a single number never could.

It's a useful way to understand something you may already have lived: an orthostatic test — standing up while your blood pressure is measured — comes back normal, and yet standing still makes you feel dizzy, foggy, or wrong in a way the number doesn't capture. One small study measuring blood flow to the brain directly, rather than just blood pressure at the arm, found that people with post-COVID symptoms and people with POTS (a related dysautonomia) both showed roughly a 20% drop in blood flow to the brain on standing, against about 3% in healthy people — even in cases where blood pressure itself looked unremarkable (Novak et al. 2022). The prediction this points to is specific: it's not that your blood pressure will fall, it's that something further downstream can go wrong while the blood-pressure cuff says everything is fine. That's a real gap between what a standard test measures and what you actually feel, and it's worth naming to a clinician in exactly those terms.

This is also why HRV — the tiny beat-to-beat variation this chapter mentioned earlier — is interesting as a relationship, not a single score. One 2026 study tracked continuous HRV in long-COVID patients over several days and found that exertion near a person's own aerobic threshold was followed by a suppressed HRV that lingered for twenty-four hours afterward, more so with harder or longer exertion (Ruijgt et al. 2026). That's one observational study, not yet independently repeated, and its own authors were careful to frame it as supporting general overexertion awareness, not a validated threshold for detecting post-exertional malaise. But it fits the larger pattern: attunement isn't about staring at one number until it crosses a line. It's about noticing how the systems that are supposed to work together are relating to each other, over time, against your own baseline.

What this means today: if a test comes back "normal" but you don't feel normal, that is not a contradiction to dismiss — current research on how these two response systems interact gives a real, honest reason why a normal reading and a real symptom can coexist. Bring both to your clinician.

3.5 The tools built for this

Two tools exist specifically to support this kind of attunement, and it's worth knowing exactly what each one is and isn't.

pacing.561.group is a free calculator, built to run locally: nothing you enter leaves your own device unless you choose to export it. It's built to hold your own rolling history — heart rate, sleep, activity, symptoms, however much or little you choose to log — and hand it back to you as a pattern, not a verdict. When you want your data, it's yours to export as a file, in full, at any time. It does not diagnose anything, and it will not tell you today is "good" or "bad" — it will show you how today compares with your own recent past and let you draw the conclusion.

PACER — Personal Agent Collecting Energy Readings — is a companion agent designed to sit alongside a tool like the calculator: picture a tree frog living in a culvert, watching the water level it actually lives in, not a lake somewhere else. PACER is specified in detail, including exactly how it would handle your data, but as of this writing it has not been built. It's named here so you know it's coming and what it's meant to do, not because you can use it yet: readings would never leave your device except as a short summary, and it would read the calculator's own exported history so your data moves between tools without you losing custody of it.

What this means today: you can start using the calculator today, for free, with your data staying on your own device. PACER is a real plan, not yet a real product — check back for it rather than looking for it now.

3.6 Keeping notes on a bad day

You do not need a perfect log. On a day when getting to the kitchen is the whole agenda, a perfect log is not a realistic ask, and this practice was never meant to add another burden to a day that already has too much in it. What you need is the minimum that lets your future self, or your clinician, see the pattern later.

That minimum is small. A resting heart rate reading, if you have a way to take one, once a day. A rough symptom note — better, same, worse than yesterday, in a word or two. What you did, in the plainest terms: "walked to the mailbox," "video call, 20 minutes," "nothing." That's it. Three lines, most days, written in whatever app, notebook, or voice memo is easiest to reach on your worst day, because the version of logging that actually happens on a bad day beats the elaborate version that only happens on a good one.

The value of even this much shows up two or three days later, when you're checking back the way the last section described. A three-line entry from Tuesday is what lets you connect Thursday's crash to something specific, instead of it just arriving as another bad day for no reason you can name.

What this means today: pick the three things you'll log — a number if you have one, a word for how you feel, a word for what you did — and log them today, even if today is a bad day. Especially if today is a bad day.

3.7 What to do today

Start with one week of noticing, before you try to change anything. Log your three lines. If you're tracking heart rate or HRV, look at it against your own last month, not against a general chart. When you do something, mark your calendar to check

back in two and three days, and actually look. If a rule of thumb like the heart-rate-plus-fifteen ceiling is useful to you, use it — as a rule of thumb, not a verdict. And if a test comes back normal while you don't feel normal, write down exactly what you felt; that gap is real information, not something to talk yourself out of.

None of this requires more energy than you have today. It requires noticing what you already have — your own history — and starting to read it as the instrument it actually is. Chapter 3 takes what Attune helps you see and turns it into the daily practice of Care.

Chapter 4

Chapter 3: Care

Pause set your ceiling before you feel it coming. Attune taught you to read your own history instead of a stranger's average. This chapter is about what happens in the space those two verbs protect — the actual hours of being sick, and the people, tools, and conversations that make those hours more bearable. Care, in PACE Yourself, means one thing specifically: repairing what strain has broken, never charging you for the repair.

4.1 What this program will not do

A lot of what gets sold to people with a chronic illness is built to need you back tomorrow. A streak you don't want to break. A guilt nudge when you miss a day. A subscription that quietly assumes you'll keep paying whether or not it's helping. A confident diagnosis from a wearable that, if you read the actual research behind it, is one small study away from being an open question. None of that is in this program, on purpose.

Streaks and guilt are the sharpest problem, because of what you already know from Pause: the cost of overdoing it arrives late and lands hard. A habit tracker that makes you feel bad about a missed day is asking for exactly the thing pacing exists to prevent — pushing past today's real capacity to protect a number on a screen. A program built for post-exertional malaise cannot also be built to punish rest. And a confident-sounding number — a score, a percentage, a "you're at risk" flag — sold as more certain than the evidence behind it actually is, is its own kind of harm: it takes a genuinely open question and hands it to you as settled, which either scares you into overcaution or, when it turns out to be wrong, teaches you to stop trusting anything the tool tells you.

There's a plainer way to say why this matters, underneath the specific rules. Real protection is something people build together and draw on together — the guideline a clinic follows because other patients' outcomes taught it something, the pacing knowledge that circulated through ME/CFS communities for decades before long COVID gave it a bigger audience, the fact that you are reading this book at all because people before you fought to get post-exertional malaise taken seriously. Protection that instead gets sold to you as something only you can buy, one subscription at a time, walling you off rather than connecting you to what

others have already learned, is a thinner and lonelier kind of safety. Care, here, means staying on the first kind.

4.2 Caring for the body: the orthostatic conversation

If standing up makes you dizzy, lightheaded, or foggy, and a clinic has told you your blood pressure was normal, you are not imagining the mismatch. One of the house's own working explanations for post-exertional malaise proposes that the two arms of your body's stress response — the fast nervous-system arm and the slower hormone arm — stop coordinating with each other in this illness. If that is right, the specific, testable prediction it makes is not that your blood pressure drops. It's that blood flow to your brain can fail while your blood pressure holds steady, because the vessels aren't responding normally to the signals telling them to constrict. That is exactly the presentation that gets people told their exam was normal. This is a well-supported explanation, not yet a proven one — flag it as reasoning from mechanism, not as an established diagnosis.

Whether or not that specific explanation turns out to be right, orthostatic intolerance itself is common and real in this population. A postural heart-rate jump on standing — postural orthostatic tachycardia syndrome, or POTS — shows up in roughly one in a hundred people at some point after any infection, and in around three in ten people already sick enough with long COVID to be seen at a specialty clinic (Yong et al. 2023; Björnson et al. 2025). Separately, small-fiber neuropathy — damage to the finest nerves, the ones that help control blood vessel tone — turns up on skin biopsy at strikingly elevated rates in post-COVID patients compared to healthy people (Donadio et al. 2024). One small, unblinded study found people with this neuropathy improved more often after treatment with intravenous immunoglobulin than people who went untreated, though the study's authors were explicit about how far that particular signal is from proof (Dalakas 2024).

What actually helps, in practice: compression on your legs and abdomen, worn before you stand rather than after you're already dizzy. Deliberate positional strategies — sitting on the edge of the bed before standing, crossing your legs and tensing your calves while upright, raising the head of your bed a few inches at night, avoiding long stretches of still standing (a shower chair is not giving up, it's compression's cousin). Fluids and salt, increased under a clinician's guidance rather than guessed at. And medication — the drugs clinicians reach for here (beta-blockers, midodrine, ivabradine among them) only with a clinician who knows your case, never self-started. The best current trial evidence on this is worth knowing before that conversation: a large platform trial testing one such drug against placebo for long-COVID POTS found it did what it was supposed to do to the heart rate, but that alone didn't move people's symptoms — what did move the needle

was that same drug paired with structured, coordinated non-drug support (Fudim et al. 2026; RECOVER Initiative, National Institutes of Health 2026). The drug and the support work as a pair, not as substitutes for each other.

4.3 Sleep that doesn't refresh

Unrefreshing sleep is not a side complaint in this illness — it's one leg of the same three-legged diagnostic stool as post-exertional malaise and the fatigue itself (Institute of Medicine Committee on the Diagnostic Criteria for Myalgic Encephalomyelitis/Chronic Fatigue Syndrome 2015). You can sleep eight or nine hours and wake up as tired as when you lay down. That is not a discipline failure and it does not mean you're sleeping "wrong." It means treating sleep quantity as the whole goal will disappoint you; the honest aim on a bad week is protecting the conditions around sleep — dark, quiet, a consistent wind-down, a bedroom that isn't also your workstation — rather than expecting a fixed number of hours to fix how you feel the next day.

4.4 Hydration, electrolytes, and temperature — labeled honestly

Extra fluids and salt (electrolytes) are standard first-line advice for orthostatic intolerance generally, extended to long COVID by clinical practice rather than proven in a long-COVID-specific trial — that is a reasonable extension from the broader POTS literature, not yet its own tested result in this population, and it should be offered to you with exactly that caveat. The same honesty applies to heat: POTS symptoms are characteristically worse in warm environments and improve lying down (Grubb 2015), which is a documented feature of the diagnosis, not a promise that any particular cooling strategy will fix a given day. Given that, a cooling towel or a fan by the bed on a hot day is a reasonable comfort measure, offered as exactly that — comfort, not treatment.

4.5 Sensory load and the rooms you rest in

Cognitive exertion counts as exertion, and so does sensory load — bright light, noise, a cluttered visual field, a phone buzzing every few minutes. None of that is separately proven to trigger post-exertional malaise the way physical exertion is documented to, but it costs the same kind of attention and arousal your body is already short on, so the practical move is the same: build a low-stimulus space you can retreat to before you're in a flare, not after. An eye mask and a pair of

earplugs are unglamorous, but they turn “I need to lie down somewhere dark and quiet” from a scramble into something you already have ready.

4.6 The clinician conversation

Walking into an appointment with a clinician who doesn’t know this territory well, the most useful thing you can say plainly is: normal results here don’t rule out what you’re describing. A standard blood-pressure check can miss the cerebral-perfusion pattern above entirely, because it’s designed to. Bring your own data if you track it — a printed pacing log, or an exported history from a free tool like the pacing calculator at pacing.561.group — so the conversation starts from your own pattern instead of your memory of a bad week. Ask directly whether pacing, not a fixed exercise schedule, is what they’d recommend; UK national guidance explicitly recommends pacing over graded, schedule-based exercise programs for this illness (National Institute for Health and Care Excellence 2021). It’s also fair to ask how the clinic hears about a recommendation that made someone worse — one survey of specialist clinics found none had a systematic way of tracking treatment-related harm, and none had recorded a single case of it despite people dropping out of care (McPhee et al. 2019). A clinician who can sit with “that’s a real, disproportionate, delayed reaction, not you being dramatic” (Stussman et al. 2020) is one worth keeping.

4.7 Caring for a person who has this

If someone you love has post-exertional malaise, the single most useful thing you can internalize is that the second day — not today — is the actual test of whether today went okay. A visit, a conversation, even a good day out, can look fine in the moment and cost three days later. Plan for that instead of being surprised by it: check in the day after, not just the day of. Help that costs the person nothing to receive — dropping off food instead of hosting, a phone call kept short and endable instead of a long visit, running an errand without needing to be thanked for it in person — is worth more than help that requires them to perform wellness for you while receiving it. And when a plan needs to change because of a flare, treating that as normal and expected, not as a disappointment they have to apologize for, is itself a form of care.

4.8 The Inflammation Station kit

Culvert Health also puts together a small physical kit of comfort items, chosen by exactly the symptom logic in this chapter: a soft stretch wristband, a contoured eye mask and a pair of foam earplugs for the low-stimulus room, an electrolyte drink mix, a printed pacing card and a symptom journal for tracking your own pattern, a cooling towel for hot days, a neck pillow, and a card linking to the free ebook and the pacing calculator. None of these nine items is a medical device, and nothing in the kit is sold with a diagnostic, therapeutic, or treatment claim — it exists to make the ordinary comfort measures in this chapter easier to have on hand, nothing more.

4.9 What to do today

Pick one thing from this chapter that costs you nothing to try this week: raise the head of your bed a couple of inches, put an eye mask by your pillow, write down what today actually cost you so you have real data for your next clinician visit instead of a guess. Care is not a bigger ask than that. It's the ordinary, unglamorous work of making today survivable — and it is never going to ask you to push through to earn it.

This is not medical advice; talk with a clinician who knows your case before changing medication, fluids, or salt intake.

Chapter 5

Chapter 4: Engage

5.1 What “engage” does not mean

If you have spent any time around chronic-illness advice, “engage” probably makes you brace. It sounds like the start of a pep talk: get back out there, push a little further each week, don’t let the illness win. That is not what this chapter means, and the difference is not a matter of tone. It is a difference in mechanism.

Book 1 walked through why post-exertional malaise breaks the ordinary rule of “push a bit, rest, repeat”: the flare from an exertion doesn’t show up right away. It can take 12 to 72 hours to arrive, and once it does, it can last days to weeks, not hours (Stussman et al. 2020). That delay is the whole problem with a calendar-based plan. A schedule that says “walk five minutes longer this week than last” is written before your body has reported back on this week, because it can’t report back that fast. By the time the flare from Tuesday’s push lands on Thursday, the schedule has already told you to push again on Wednesday.

That specific plan — fixed increases in activity on a predetermined timetable, regardless of how you respond — has a name: graded exercise therapy. It is a real, evidence-supported treatment for real conditions, mainly ones where deconditioning is the actual problem. It is also the approach the 2011 PACE trial tested and recommended for ME/CFS, using methods and recovery thresholds that the UK’s National Institute for Health and Care Excellence later found unreliable enough to withdraw its own earlier recommendation over — no post-exertional-malaise measurement, outcome goalposts moved mid-trial, “recovery” thresholds that overlapped with the criteria for getting into the trial in the first place. NICE’s 2021 guideline replaced that recommendation with pacing instead, for exactly the population this book is written for (National Institute for Health and Care Excellence 2021). This isn’t a footnote about one flawed study. It is why this brand is named PACE Yourself in the first place: a deliberate reclaiming of a word that trial’s approach never actually did.

Patients have told researchers directly what happens when the graded-exercise approach meets a body that runs on post-exertional malaise instead of ordinary deconditioning. In one large patient-reported-outcome survey, more than half of respondents who had tried graded exercise therapy reported that it made them substantially worse (Geraghty, Hann, and Kurtev 2017). And when researchers

went looking for how well that harm gets tracked, they found a gap: a freedom-of-information survey of NHS ME/CFS specialist clinics in England found none had a systematic protocol for recording treatment-related harm, even though every clinic acknowledged patients dropping out of treatment — which means harm may be going unrecorded rather than not happening (McPhee et al. 2019). Put those two findings together and the honest picture is: patients report real, common harm from calendar-based pushing, and the systems meant to catch that harm mostly aren't looking.

So here is what engagement actually means in this program: you advance when your body has told you the last exertion is fully paid off, not when a calendar says a week has passed. Progress is real, but it is symptom-contingent and reversible — if a relapse happens, you step back, not through it. There is no finish line where you graduate off pacing into "normal" activity on a fixed timetable. This is the mechanistic opposite of graded exercise therapy, not a gentler version of it.

What this means today: if a plan for your week says "more than last week" without asking how last week actually went, that plan is asking the wrong question. The right question is always: has the cost of what I already did been paid off yet?

5.2 Choosing what to spend on

Once pushing-through and calendar progression are off the table, the practical question becomes: given a fixed and often small amount of energy, what do you spend it on?

This is where the "energy envelope" from earlier in this program stops being an abstraction and becomes a daily decision. Book 1 explained the mechanism behind why the envelope has to be respected even on a day you feel fine: aerobic capacity behaves like a rate, not a reserve you draw down and refill, so the failure point shows up at low intensity rather than after you've used up some stockpile. There is no bank of extra energy sitting behind a good day that you're free to spend just because you have it. The envelope on a good day is still the envelope.

Inside that envelope, though, you get to choose. This is usually called energy accounting or energy budgeting, and the comparison to money is useful because it points at the right question. A financial budget doesn't ask "how do I spend nothing," it asks "given what I have, what actually matters enough to spend it on." The same is true here. If today's envelope holds room for one significant thing, the question isn't whether to spend it — you're going to, one way or another, even by accident — the question is whether you spent it on the thing you'd have chosen if you'd been asked in advance.

This is also where pacing stops being purely defensive. The point was never to do as little as possible; it was to avoid the crashes that make the bad months,

so that what you do have goes toward what you value. Some days that's a task that has to happen. Some days it's a phone call with someone you love instead of the laundry. Both are legitimate spending. What makes it a budget rather than a guess is writing down, even loosely, what a task actually costs you — and noticing, over weeks, which categories of activity reliably cost more than they look like they should from the outside. Cognitive effort, social effort, and standing upright all draw on the same limited account, even when only one of them looks like "activity."

What this means today: before you spend today's envelope, ask which one thing you'd protect if you could only protect one — and protect that one first, before the day fills up with things that arrived by accident.

5.3 Work, study, and relationships inside the envelope

The energy-accounting frame gets harder, not easier, once other people are involved — a job, a class, a family that has its own schedule. Nobody else's calendar was built with a 12-to-72-hour delayed flare in mind, which means the honest version of engaging with work, study, or relationships usually requires you to say something out loud that the people around you did not expect to hear.

The core adjustment is the same one from the rest of this chapter, translated into someone else's terms: build in recovery time before you're asked to perform again, not after you've already crashed. A workday that ends at five and expects you fresh at nine the next morning is, for many people with post-exertional malaise, not actually asking for one day of work — it's asking for the cost of that day to already be paid off by morning, which the 12-to-72-hour delay (Stussman et al. 2020) makes an unreasonable bet more often than not. The honest ask, whether to an employer, a professor, or a partner, is rarely "less work." It's usually "the same work, spread differently" — a compressed schedule, an asynchronous deadline instead of a live one, a standing understanding that a bad day means a shifted day, not a missed obligation.

This is uncomfortable to ask for, partly because it requires explaining a delay most people have never had to think about. It helps to say the mechanism plainly rather than just the request: "the cost of today doesn't show up today — if I overdo it now, you'll see it Thursday, not this afternoon." That sentence does real work. It reframes what looks like flakiness or inconsistency as a predictable, plannable fact about how your body settles its accounts, not a character trait.

Relationships carry their own version of this. Emotional exertion spends from the same envelope as physical exertion — a hard conversation can cost as much as a walk you didn't take. People who care about you can usually handle knowing that in advance far better than they can handle watching you disappear for a week afterward with no explanation. Telling someone "that conversation mattered to me

and it's also going to cost me the next two days" is not a burden you're putting on them. It's the information they need to actually be there for the version of you that shows up on day three.

What this means today: pick one place — work, school, or one relationship — where you have not yet said the delay out loud, and say it once, plainly, as a fact about timing rather than an apology.

5.4 Engage the evidence

Book 1's final chapter taught you how to read a study yourself: check the case definition, check the sample size, check whether anyone else has replicated it, read the confidence interval instead of just the headline. That skill doesn't get re-taught here — go back to that chapter if you want the mechanics again. What belongs in this chapter is what to do with that skill once you have it: use it, on purpose, as part of how you engage with your own illness.

Research on infection-associated illness moves in a loop, and understanding the loop changes how you read a new headline. Patient-reported experience — what actually happens to people pacing in the real world — is what motivates researchers to run small trials in the first place. Those small trials get pooled into meta-analyses, which tell you, honestly, how much confidence the field has actually earned so far. The most recent pacing meta-analysis pooled just five studies and found a large point estimate for fatigue improvement that still fell short of statistical significance, alongside smaller effects on pain and function the authors themselves called trivial (Sanal-Hayes et al. 2024). That is not a discouraging result to read — it's the loop working exactly as it should: patient experience generated a hypothesis worth testing, a handful of small trials tested it, and the honest state right now is "promising, not yet proven," which is precisely what the study's own authors said.

You can watch this loop happen in real time with the two-day exercise-test marker discussed in Book 1. One study found a measurable decline on day two (Keller, Pryor, and Giloteaux 2014), a larger controlled study confirmed it (Snell et al. 2013), and then a newer, larger 2026 study did not replicate it (Mancini et al. 2026). That is not researchers failing each other. That is the loop mid-cycle, with the next study yet to be run. Watching a live disagreement like this, instead of only reading settled textbook facts, is part of what it means to engage the evidence as it actually exists.

The most current example of the loop for this book's own subject is RECOVER-AUTONOMIC, a large NIH-funded platform trial testing management approaches for long-COVID dysautonomia. Its ivabradine arm — 181 participants, randomized, placebo-controlled — reported a secondary finding worth knowing: patients

who received coordinated non-drug care (which includes activity pacing guidance) alongside their assigned treatment reported greater symptom improvement than those on medication management alone (RECOVER Initiative, National Institutes of Health 2026). That result has, as of this writing, been presented at a cardiology conference but has not yet appeared in a peer-reviewed journal publication — the trial's design and rationale are published (Fudim et al. 2026), but the results themselves are still preliminary, conference-only evidence, not yet closed out through peer review. It is genuinely encouraging and it is not yet the last word.

What this means today: when you see a new headline about pacing, long COVID, or ME/CFS, ask the same four questions Book 1 taught you before you let it change how you feel: who was studied, how many, has anyone else confirmed it, and is this preliminary or established.

5.5 Engage community and advocacy

The honest state of the pacing evidence — real, promising, and still thin — is not a private fact for you to sit with alone. It's an argument for what should get funded next, and you are allowed to make that argument.

Trials specifically designed around post-exertional malaise, with PEM measured directly rather than assumed, and with symptom-contingent designs instead of fixed schedules, are overdue. The gap is not subtle: the field has one meta-analysis pooling five small pacing studies (Sanal-Hayes et al. 2024), one contested two-day exercise-test marker with studies pointing in opposite directions (Keller, Pryor, and Giloteaux 2014; Snell et al. 2013; Mancini et al. 2026), and a documented absence of systematic harm-tracking at the clinics that patients actually attend (McPhee et al. 2019). That is not enough evidence to run a fifteen-year-old field on, and it is not an accident that it's thin — post-exertional malaise has historically been under-recognized outside ME/CFS communities, and clinical trials in this space are expensive and only recently well-funded at all.

What to ask for, concretely, when you have the energy for advocacy — a public comment to a funding body, a conversation with a clinician-researcher, a note to a patient organization — is specific rather than general: trials that measure post-exertional malaise as a primary outcome, not an afterthought; designs that let treatment adapt to a participant's actual response instead of running on a fixed schedule; and routine, structured harm reporting at the clinics doing this work, so that what happened to the patients in the McPhee survey (McPhee et al. 2019) stops being invisible. None of that requires you to become a research advocate full time. It requires knowing what a good trial for this condition would look like, so that when you do have a moment of energy to spend on it, you spend it asking for the right thing.

What this means today: you don't owe the research field your energy. If you have some to spare, spend it asking for one specific thing — PEM measured directly, or harm tracked routinely — rather than a general "more research."

5.6 Engage the tools

Part of engaging on your own terms is keeping your own data where you can actually use it. The free pacing calculator at pacing.561.group computes over the readings you enter and says nothing you didn't ask it to; your history there can be exported and re-imported as your own file, on your own terms, at any time. It is yours — not locked behind an account, not something you have to request back.

That same exported history is built to travel. The identical file you export can be handed to a clinician, or fed into other tools built to read it, without you re-entering a single number. A paid, agent-facing version of the same pacing compute exists specifically for that use — for clinicians and other tools that want to query the analysis programmatically — but it is a service other systems can call on your behalf, never something you are required to use yourself. The free, human-facing calculator is the whole tool for a person managing their own pacing; the paid option exists downstream of it, for people building on top of your data with your say-so, not instead of it.

A companion agent, PACER, is specified to extend this same approach — a daily check-in that collects your readings, keeps you company inside your energy limit, and imports the same export file the calculator already produces, so your history moves between tools without you losing custody of it. As of this writing it is specified but not yet built; when it exists, it will follow the same rule as everything else in this section: your readings stay on your device unless you choose otherwise.

What this means today: if you're tracking anything about your pacing right now — on paper, in an app, in your head — ask whether you could get that history out in a form you could hand to someone else, unprompted, if you ever needed to. If the answer is no, that's worth fixing before you need it.

5.7 What to do today

Engagement, in this program, is not a reward you earn for getting better. It's the ordinary shape of a day and a week lived inside your envelope: you do the things that matter to you, you plan for the delayed cost rather than being surprised by it, you say the mechanism out loud to the people who need to understand it, and you let the evidence — not a calendar — tell you when to do more.

Concretely, today: pick the one thing in your day you'd protect first if the envelope was tight. If there's a place — work, school, one relationship — where

the delayed cost of exertion has never been said out loud, say it once, plainly. If you're not sure your own tracking could be handed to someone else tomorrow if you needed it to be, take the five minutes to check. And the next time a headline about this illness crosses your feed, run it through the same four questions before you let it change how you feel about today.

None of this promises a cure or a date by which you'll be doing more than you are now. What it offers instead is a way of moving that doesn't cost you the next week to get through this one.

Chapter 6

Chapter 5: The Loop and the Second Day

You have now met the four verbs one at a time: Pause, Attune, Care, Engage. This chapter puts them back together — not as four separate skills you switch between, but as a loop: one order a day moves through, and one longer rhythm a week and a month move through around it. If the earlier chapters were the parts, this one is the machine.

6.1 The order of a day

A day in this program starts before anything happens. Pause comes first, on purpose: you set your ceiling before you feel anything that would tell you where it is. A post-exertional-malaise flare can arrive 12 to 72 hours after the exertion that caused it, and last days to weeks once it does (Stussman et al. 2020). By the time your body signals “that was too much,” the bill for it is already in the mail. So the ceiling gets set from your own history, not from how you feel this minute.

Attune comes next: the morning read. How did you sleep, what does today’s resting heart rate look like against your recent pattern, how does your body feel against yesterday. Nothing here is a test you pass or fail — it’s a description, today looks low, ordinary, or better than the last few. Whatever tool you use, from a paper log to a wearable to the pacing calculator, its job is to hand you back your own data, not a verdict.

Care is what you do with that read: building the day’s plan to fit inside today’s envelope, not the one you wish you had. If today’s read is low, the plan for meals, hydration, rest breaks, and sensory load shrinks with it — along with anything specific to your own management, from position if standing worsens you to protecting the sleep window to keeping noise and light down.

Engage is the activity itself — physical, cognitive, or social — chosen from inside the envelope Attune and Care just set, never from a schedule written last week. If the plan said “grocery trip” but this morning’s read says low, the trip moves.

Then the loop closes: Pause again. Not dramatic, just the same habit repeated before the next block, the next errand, the next conversation that might run long — continuously, in miniature, all day, with the morning version setting the frame.

What this means today: Pause before you decide, Attune before you plan, Care while you build the day, Engage inside what you built, Pause again. Skipping straight to Engage — deciding what to do before checking what today can hold — is the single most common way the loop breaks.

6.2 The second day as the audit

Because the flare from an exertion can arrive as late as three days afterward — the 12-to-72-hour window from earlier in this chapter — yesterday is not actually done yet when you wake up today. The second day is where the loop checks its own work — a simple habit, not a test. Each morning, note two things: what you did yesterday, and how you feel today. Over time this becomes a short, honest record — not of willpower, but of what your body actually did with what you asked of it, and the only data that can tell you where your real envelope sits, because nobody else has access to it.

If today is a flare — worse than an ordinary bad day, following an exertion from somewhere in that window — that is not a failure of the plan. It is the plan working exactly as intended: catching information the felt sense in the moment could not have given you, because that delay is exactly why pacing exists as a discipline rather than a feeling. The scientific record on the mechanism behind the delay is itself still being worked out — a repeated exercise test given a day apart has shown a measurable decline on the second day in some studies (Keller, Pryor, and Giloteaux 2014; Snell et al. 2013) and no such decline in a larger, more recent one (Mancini et al. 2026) — but the lived pattern the test was trying to capture, a delayed and disproportionate crash, is well documented regardless of which laboratory marker eventually explains it (Stussman et al. 2020).

When a flare shows up, the response is to re-estimate, not to punish. Whatever activity preceded it becomes a data point: that was more than today's body could absorb. The envelope shrinks around that information, at least for a while. There is no shame accounting here and no streak to protect — a flare is information, and gathering information is the entire point of the record.

What this means today: the morning after any day with real activity in it, take thirty seconds to write down what you did and how you feel. If a flare comes, don't fight it — rest into it, and let it tell you something about where the line actually was.

6.3 The weekly review

Daily audits are the input; the weekly review is where they add up to something. Once a week, look back over the log: how many days were stable, how many were

flares, what preceded each. Re-estimate your envelope for the coming week from that — not from a guess or from how ambitious you feel on review day, but from what the last seven days actually showed you.

Some things stay fixed week to week: the general approach (pause first, plan from the morning read, log the outcome), any hard limits you've clearly identified, the daily habits around sleep and hydration that support the whole loop. What moves is the size of the envelope, and sometimes its shape — cognitive load may matter more than physical load this month, or mornings may prove more reliable than evenings. The review is where this month's actual data updates last month's assumptions.

What this means today: pick one day a week — the same day each time is easiest — to sit with your log for a few minutes and ask two questions: which days were stable, and what did they have in common? Let the answer, not your hopes for the week ahead, set next week's ceiling.

6.4 What a good month looks like

It is worth saying plainly what success looks like here, because it is not what most health advice trains you to expect. A good month is not a month where you did more — it is a month with fewer crashes than the last, more stable days, shorter and milder flares when they do happen, and your envelope getting clearer rather than murkier. If activity stayed the same but crash frequency dropped, that is real progress. If activity went up because the envelope genuinely widened — which some patients and clinicians report, though this isn't something the record here can confirm — that shows up as fewer crashes at the new level too, not as grit against how you feel.

This is a different scoreboard than the one you're probably used to from gyms or productivity culture. The scoreboard here is stability, not output.

What this means today: if you find yourself judging a week by how much you got done, try judging it instead by how many of its days were stable. That's the number this program is actually trying to move.

6.5 When to change the plan, and when not to

Change the plan when the log tells you to. A pattern of flares following a specific kind of activity, several stable weeks that suggest room to test a slightly larger envelope, a new symptom needing fresh calibration — these are grounded in your own data, exactly what the weekly review is for.

Do not change the plan because of a single good day, because you feel behind on things you used to do, because someone else with a similar condition is doing

more, or because a fixed calendar says it's time to progress. None of that is your data. The evidence we do have on the opposite approach — increasing activity on a predetermined schedule regardless of how the body responds, known as graded exercise therapy — points the same direction: a patient-reported-outcome survey found large negative responses to it are common (Geraghty, Hann, and Kurtev 2017), a clinic freedom-of-information survey found that clinics using it have not systematically tracked whether it caused harm (McPhee et al. 2019), and the UK's National Institute for Health and Care Excellence, in its own guideline review, set aside its use for myalgic encephalomyelitis because the trial evidence behind it did not hold up to scrutiny (National Institute for Health and Care Excellence 2021). The evidence for pacing itself is not yet as strong as anyone would like — a 2024 meta-analysis of the small number of controlled pacing studies found a real but not statistically significant benefit for fatigue, and only a trivial one for physical function and pain (Sanal-Hayes et al. 2024) — but the case against the schedule-driven alternative is clear enough that this program does not use one.

What this means today: before you decide to push your envelope wider, ask whether the last two or three weeks of your own log actually support it. If they do, expand carefully and watch the next second day closely. If the only evidence is impatience, wait.

6.6 The program never graduates you

There is no final chapter where the pacing stops. This program does not build toward a day when you no longer need to Pause, Attune, Care, and Engage — it is not a course with an end date. Patients who have practiced pacing over years consistently describe it as the thing that prevents crashes, not as a phase they moved through and past (Stussman et al. 2020; Jason and Dorri 2022). If your envelope widens over time, the loop widens with it — you're still checking, still logging, still reviewing. The goal was never activity for its own sake; it was staying inside whatever your envelope actually is, for as long as that takes.

6.7 The four verbs, one page

Keep this page. Copy it, print it, put it wherever you'll actually see it on a hard morning.

Pause. Stop before the limit, not at it. Set today's ceiling from your own history, before you feel anything that would tell you where the line is.

Attune. Read this morning honestly — sleep, resting readings, how your body feels against your own recent pattern. Describe what you find. Never treat any single number as a diagnosis or a prediction.

Care. Build today's plan to fit inside today's envelope. Hydration, rest, position, sensory load, sleep — all of it counts as part of the plan, not just the activity.

Engage. Do the chosen activity inside the envelope you just set, on symptom-contingent terms. If today doesn't hold what yesterday's plan expected, the plan moves, not the envelope.

The second day. The day after any real activity is an audit day. Log what you did and how you feel. A flare is information about where the line was, not a failure.

The week. Once a week, review the log. Let what actually happened set next week's envelope — not hope, not impatience, not a calendar.

6.8 A closing note on honesty

Some of what this chapter rests on is well established: post-exertional malaise's delayed, disproportionate timing is documented and consistent across patient accounts (Stussman et al. 2020), and the case against progressing activity on a fixed schedule, regardless of symptom response, is about as clear as evidence in this field gets (Geraghty, Hann, and Kurtev 2017; McPhee et al. 2019; National Institute for Health and Care Excellence 2021). Other parts are reasoned from mechanism and patient experience rather than from controlled trials — the pacing strategy itself has real but still-limited controlled evidence behind it (Sanal-Hayes et al. 2024), and the exact biological marker of the second-day flare remains an open, contested question in the research literature even while the pattern patients describe is not (Keller, Pryor, and Giloteaux 2014; Snell et al. 2013; Mancini et al. 2026). This book has tried to say, every time, which kind of ground a claim stands on.

What this program cannot promise: a cure, a recovery timeline, or that pacing alone will fix what's wrong. What it can offer is a way to live with real uncertainty without making it worse — a loop that respects what your body has already told you, and a habit of listening for what it tells you next. This is not medical advice; it does not replace a clinician who knows your case.

Culvert Health · a division of Causeway Services · research by Achromat (Lenticule Science) · systems by Ampoule (Red Cup Engineering) · care by Splint (Coesite Operations)

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